

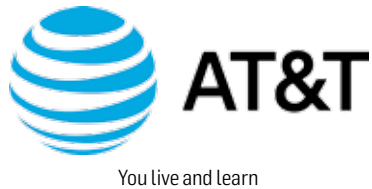
in the requirement analysis phase which can then be transformed into a highly granular design model might have averted the situation altogether.

The additional effort in granular and exhaustive definition of requirements is thus clearly worth the effort if one is trying to do something as monumental as check the weather on Mars. A comprehensive suite of test cases that cover all edge scenarios might also have detected this flaw before it ever reached production.

Case 2: AT&T communications go down

While at first glance the AT&T fiasco might look like a technical snag or a quality issue, it is also a result of design flaw within the application. The race situation that caused the self-propagating disaster resulting in the shutdown of half of the switches in USA should have been identified as a concurrency point right at the design phase, which could then ideally have been handled in a desired manner using locks.

Designing an application structure to granular detail, including the technical stack and interaction diagram is key to developing it.



As mentioned in the best practices for design chapter, a good and effective design diagram can automatically detect issues like synchronization or unnecessary components.

These can then be handled via workarounds of design without writing a single line of code yet!

Case 3: Passenger plane shot down by USS Vincennes

Perhaps the most unfortunate of all the cases, a civilian plane was shot down by the US Navy in 1988 mainly due to a poorly designed User Interface. The UI of the Aegis system was too cluttered and misleading to deduce information rapidly and hence prone to misinterpretation as seen in this case. A clear and well designed UI is very critical in retrieving and delivering information to users. Often low prioritized, the quality of the UI component specifically for customer facing applications can literally make or break them.

While we see here that the information collected by the Aegis system was 100% correct, representing it in a poor manner resulted in the operator misunderstanding the content, which in turn cascaded into this disaster.